Linear Search

- It is a basic and simple search algorithm.
- In sequential search, we compare the target value with all other elements given in the list.

→ Eg:-

	0	1	2	3	4	5
arr =	18,	12,	19,	7,	29,	50

(unsorted array)

target = 77 start

- In above example, the target value is compared with all the elements in array in sequential / linear array.

Time complexity

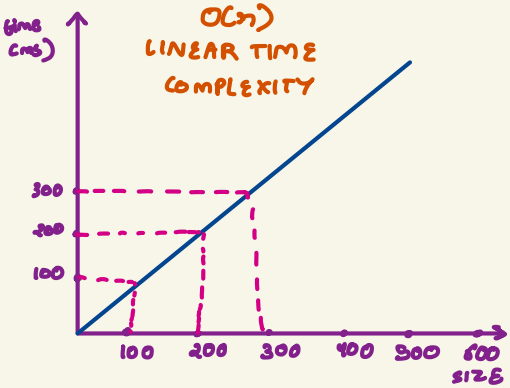
→ **BEST CASE** : $O(1)$ → constant

The best case scenario will be that the target element is situated at the zeroth index itself. Therefore, it will take only one step for the searching algorithm to find the target element.

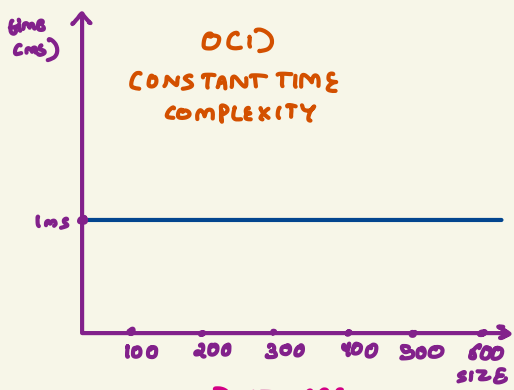
→ **WORST CASE** : $O(n)$

The worst case scenario goes through every element and still doesn't find the element.

SIZE OF ARRAY	NO. OF COMPARISONS	TIME (ms)
100	100	100 ms
200	200	200 ms
n	n	n ms



WORST CASE



BEST CASE